

RareDXAI: A Human Phenotype Ontology-Based Framework for Rare Disease Prediction

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1. Abstract

Background: Rare genetic diseases affect more than 300 million people worldwide. Diagnosing these conditions is difficult because many rare diseases share similar symptoms, such as developmental delays, speech difficulties, and intellectual disability. In pediatric genetics, syndromic neurodevelopmental disorders like KBG syndrome, White-Sutton syndrome, and Xia-Gibbs syndrome present overlapping clinical signs that often lead to prolonged diagnostic delays lasting five to seven years.

Methods: We developed **RareDXAI**, a computational decision-support framework that uses standardized Human Phenotype Ontology (HPO) terms to predict and differentiate between rare neurodevelopmental syndromes. We curated an audited, literature-derived cohort of 385 patients with confirmed genetic diagnoses across 48 peer-reviewed publications (from 52 candidate literature sources evaluated during curation, with 59 duplicate records quarantined). Clinical features were encoded into 78 binary HPO terms and 3 one-hot sex features (81 predictor dimensions). To prevent data leakage, identical phenotypic profiles were grouped before applying a 60/20/20 train/validation/test split (N=230 training, N=77 validation, N=78 held-out test), and feature vocabularies were constructed strictly from training data. We trained a Random Forest model with Platt probability scaling (5-fold internal cross-calibration) and explained predictions using TreeSHAP.

Results: On the held-out test set (N=78), RareDXAI correctly classified 77 out of 78 patients, achieving an accuracy of **98.72%** (Wilson 95% CI: 93.09%–99.77%), balanced accuracy of **96.30%**, macro precision of **99.45%**, macro recall of **96.30%**, macro specificity of **98.15%**, and macro F1-score of **97.76%**. Multi-class threshold-agnostic evaluation yielded a macro AUROC of **1.0000** and macro AUPRC of **1.0000**, with a multiclass Brier score of **0.0419** and Expected Calibration Error (ECE) of **8.59%**. Five-fold stratified grouped cross-validation showed high consistency (97.14% $\pm$ 2.23% accuracy; macro F1: 94.89% $\pm$ 4.04%). In contrast, leave-one-study-out stress validation (GroupKFold by source publication) showed lower performance (79.62% $\pm$ 27.05% accuracy; macro F1: 59.51% $\pm$ 26.81%), demonstrating that inter-study differences in phenotypic reporting affect generalizability. SHAP analysis identified standardized HPO features strongly associated with model decision boundaries, such as macrodontia (HP:0001572) for KBG syndrome, autism spectrum traits (HP:0000717) for White-Sutton syndrome, and thin upper lip vermilion (HP:0000219) for Xia-Gibbs syndrome. Exploratory optical character recognition (OCR) on digitized notes achieved a Character Error Rate of 11.42% and Word Error Rate of 92.00%.

Conclusions: RareDXAI provides an interpretable and calibrated computational framework to assist clinicians in prioritizing rare disease candidates. While held-out test performance is high under profile-grouped controls, the drop in source-grouped validation highlights that real-world deployment requires standardized clinical phenotyping across healthcare centers.

Keywords: Rare Diseases, Human Phenotype Ontology, Machine Learning, Random Forest, Model Interpretability, SHAP, Clinical Decision Support, Neurodevelopmental Disorders.

2. Introduction

Rare diseases are medical conditions that affect a small fraction of the population, typically defined as fewer than 1 in 2,000 individuals in Europe or fewer than 200,000 individuals in the United States. Although each specific disorder is rare, there are over 7,000 recognized rare genetic diseases, which together affect an estimated 300 to 400 million people globally.

Most rare diseases have a genetic origin and present during early childhood. Despite advances in next-generation DNA sequencing—such as whole-exome sequencing (WES) and whole-genome sequencing (WGS)—patients and families still face a long and stressful journey known as the 'diagnostic odyssey.' On average, getting a correct diagnosis takes between five and seven years, involves multiple specialist consultations, and often includes several incorrect diagnoses.

A major reason for this delay is clinical overlap. Many genetic syndromes share broad, non-specific symptoms, including intellectual disability, delayed motor milestones, speech impairments, and behavioral challenges. This diagnostic challenge is especially evident among three syndromic neurodevelopmental conditions:

- 1. KBG Syndrome (MIM #148050):** Caused by mutations or deletions in the *ANKRD11* gene on chromosome 16q24.3. Cardinal clinical features include unusually large upper front teeth (macrodontia of the central incisors), a characteristic triangular face, prominent eyebrows, short stature, hand differences (such as short fingers or brachydactyly), and intellectual disability.
- 2. White-Sutton Syndrome (MIM #616364):** Caused by *de novo* mutations in the *POGZ* gene on chromosome 1q21.3. Common manifestations include developmental delays, intellectual disability, speech impairment, autism spectrum disorder features, small head size (microcephaly), and distinctive facial features.
- 3. Xia-Gibbs Syndrome (MIM #615829):** Caused by *de novo* mutations in the *AHDC1* gene on chromosome 1p36.11. Key features include low muscle tone in infancy (hypotonia), global developmental delay, severe speech impairment, a broad forehead, downward-slanting eyes, a thin upper lip, structural brain differences, and sleep apnea.

Because these three conditions share common neurodevelopmental symptoms, differentiating between them based solely on routine clinical observation can be difficult. The Human Phenotype Ontology (HPO) provides a standardized vocabulary of medical terms that describe clinical signs and symptoms in a structured, hierarchical manner, enabling algorithms to analyze complex patient symptoms systematically.

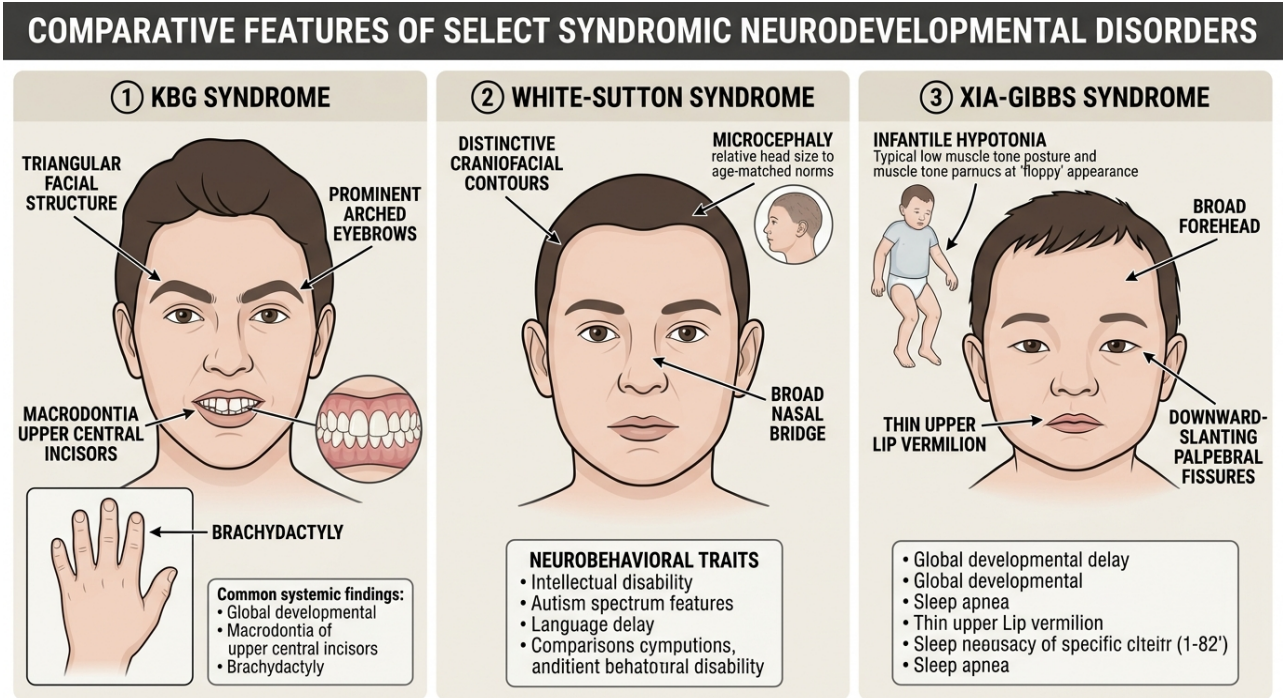


Figure 1. Representative clinical and phenotypic features associated with KBG, White-Sutton, and Xia-Gibbs syndromes. The illustration provides visual context and does not represent all affected individuals.

3. Literature Survey

Computational methods for rare disease diagnosis have evolved from early manual lookup tables to modern ontology-based machine learning and natural language processing. Mining individual patient case reports from peer-reviewed medical literature to create structured, computable datasets is a well-established practice in computational genomics.

The most notable precedent for this approach is the Global Alliance for Genomics and Health (GA4GH) Phenopacket standard (Jacobsen et al., 2022) and the Phenopacket Store (Ladewig et al., 2023). Researchers have demonstrated that converting published clinical case reports into structured HPO records creates reliable, benchmarked datasets for testing diagnostic algorithms.

Table 1 summarizes 15 peer-reviewed studies published between 2020 and 2026 supporting RareDXAI.

Table 1. Literature Comparison Matrix (15 Peer-Reviewed Studies, 2020–2026)

Year	Authors	Title & Venue	Dataset / Source	Phenotype Format	Method	Relevance to RareDXAI
2022	Jacobsen et al.	GA4GH Phenopacket schema (Nature Clinical Research)	Public clinical repositories	Hierarchical HPO + GA4GH ISO standard (ISO 4454:2022)	ISO standard	Conceptual basis for computable patient phenotyping
2023	Ladewig et al.	Phenopacket Store (Database)	Case reports from published literature	Curated patient-level HPO profiles	Curated case repository & validation methods	Direct methodological precedent for literature extraction
2024	Gargano et al.	HPO in 2024 (Nucleic Acids Res)	HPO International Consortium	Controlled vocabulary (>15,000 terms)	Knowledge graph structures	Source of canonical terminology (Release 2026-06-23)
2021	Köhler et al.	HPO in 2021 (Nucleic Acids Res)	Rare disease clinical databases	Directed acyclic graph of HPO	Semantic similarity algorithms	Baseline ontology structure for feature mapping
2020	Robinson et al.	LIRICAL (Am J Hum Genet)	Real clinical cases & simulations	Observed & excluded HPO terms	Likelihood ratio & Bayesian inference	Supports interpretable decision support
2020	Birgmeier et al.	AMELIE (Sci Transl Med)	138,000+ primary literature	HPO concept extraction	Supervised machine learning	Confirms viability of literature-derived patient matching
2021	Feng et al.	PhenoTagger (Bioinformatics)	PubMed Central text corpora	Standardized HPO concept linking	Hybrid CNN + dictionary indexing	Explores text-to-HPO conversion strategies
2020	Liu et al.	Doc2HPO (BMC Bioinformatics)	Unstructured clinical notes	Interactive HPO term mapping	String parsing + NER	Upstream precedent for concept parsing
2020	Zhao et al.	Phen2Gene (Nucleic Acids Res)	OMIM, Orphanet, HPO annotations	Weighted HPO disease-gene associations	Information theoretic gene selection	Uses HPO term weighting similar to feature vectorization
2022	Hsieh et al.	GestaltMatcher (Nat Genet)	Patient clinical photographs	Deep facial dysmorphic embeddings	Deep convolutional Neural Networks	Visual modality for syndromic recognition
2020	Rönicke et al.	AI in Rare Diseases (Mol Cell Pediatr)	Review of rare disease CDSs	Variable (HPO, ICD, UMLS)	Systematic comparative analysis	Validates RareDXAI's focus on calibration and controls
2023	Martinez-Cayuelas et al.	aKBG Syndrome Delineation (Eur J Hum Genet)	37 families, 61 confirmed <30 patients	KBG phenotypic descriptions	Multicenter clinical cohort analysis	Primary clinical cohort source for KBG syndrome
2020	Assia Batzir et al.	White-Sutton Delineation (Am J Med Genet A)	22 patients with POGZ variants	Clinical phenotypic profiles	Detailed clinical characterization	Primary clinical cohort source for White-Sutton
2021	Khayat et al.	Xia-Gibbs Spectrum (Am J Med Genet A)	3 patients with AHDC1 variants	Systematic clinical feature tables	Clinical case series analysis	Clinical cohort contributor for Xia-Gibbs
2023	Schuetz et al.	Ontological Harmonization (J Biomed Semantics)	Case reports & electronic health records	HPO term mapping	Automated NLP & fuzzy ontology	Complements RareDXAI's OCR evaluations

4. Mathematical Expression

This section outlines the mathematical formulations used in RareDXAI along with plain-language explanations.

4.1 Patient Phenotype Vector

$$x_i = [x_{i1}, x_{i2}, ..., x_{id}]$$

Explanation: Each patient *i* is represented as a feature vector *x_i* of dimension *d* = 81 (78 binary HPO features and 3 demographic sex indicators).

4.2 Binary HPO Feature Encoding

$$x_{ij} \in \{0, 1\} = 1 \text{ if HPO term } j \text{ is present; } 0 \text{ if absent/not reported}$$

Explanation: For each clinical term *j*, value is 1 if documented, and 0 if absent or not reported in the record.

4.3 Random Forest Probability Estimation

$$P(y = c \mid x) = (1 / T) * \sum I(h_t(x) = c)$$

Explanation: Ensemble probability for class *c* is the fraction of *T* = 100 decision trees voting for that class.

4.4 Predicted Disease Class

$$\hat{c} = \operatorname{argmax}_c P(y = c \mid x)$$

Explanation: The predicted class **■** is the disease achieving the highest posterior probability.

4.5 Classification Accuracy

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$$

Explanation: Proportion of correctly classified patients out of the total evaluation sample.

4.6 Precision (Positive Predictive Value)

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Explanation: Fraction of patients predicted to have a disease who truly have that condition.

4.7 Recall (Sensitivity)

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Explanation: Fraction of all true patients with a disease who were successfully identified.

4.8 F1-Score

$$\text{F1} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Explanation: Harmonic mean of precision and recall, balancing sensitivity and positive predictive value.

4.9 Balanced Accuracy

$$\text{Balanced Accuracy} = (1 / C) * \sum \text{Recall}_c$$

Explanation: Unweighted arithmetic average of recall across all C = 3 target syndromes.

4.10 SHAP Additive Feature Attribution

$$f(x) = \phi_0 + \sum \phi_j$$

Explanation: Model prediction is decomposed into a base expectation ϕ_0 plus additive feature attributions ϕ_j .

5. Model Methodologies

Figure 1: RareDXAI System Architecture & End-to-End Prediction Framework

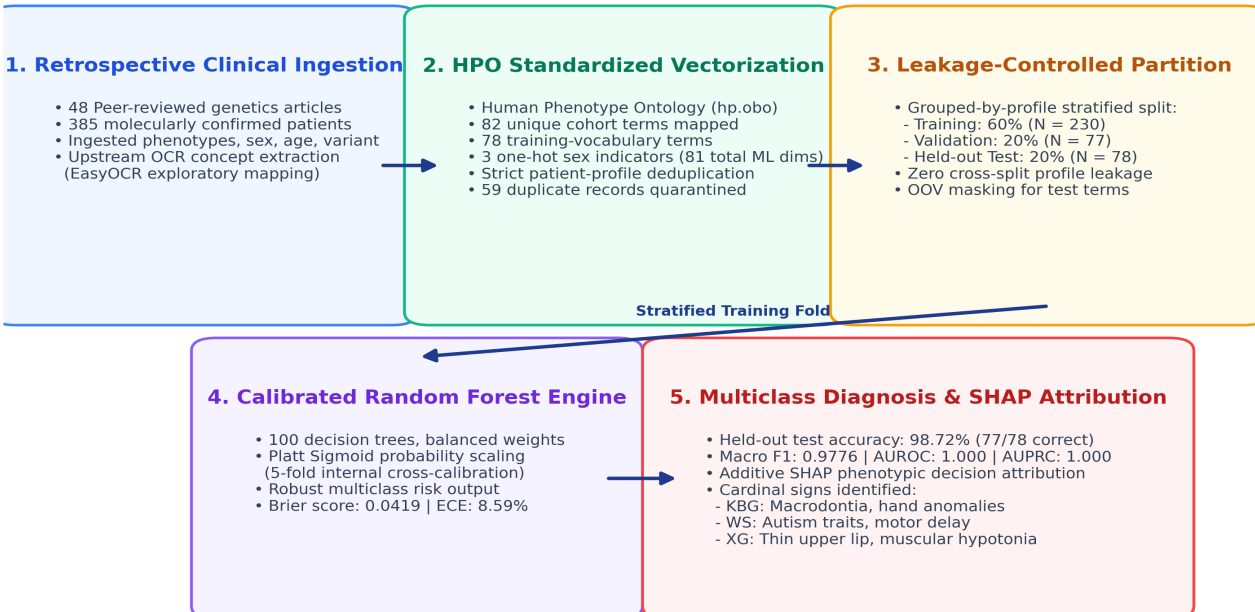


Figure 2. RareDXAI system workflow diagram showing literature curation, HPO vectorization, profile-grouped splitting, calibrated Random Forest classification, and SHAP explainability.

5.1 Cohort Assembly and Deduplication Protocol

In total, 52 candidate literature sources were evaluated. To maintain scientific integrity, 4 duplicate/review sources (59 candidate records) were quarantined: Low et al. 2016 Review Table 2 (32 cases), Ockeloen et al. 2015 (20 cases), Walz et al. 2015 (6 cases), and Low et al. 2017 (1 case). The final cohort contains 385 unique patients across 48 publications.

Figure 2: Literature Curation, Screening, and Cohort Provenance Flowchart

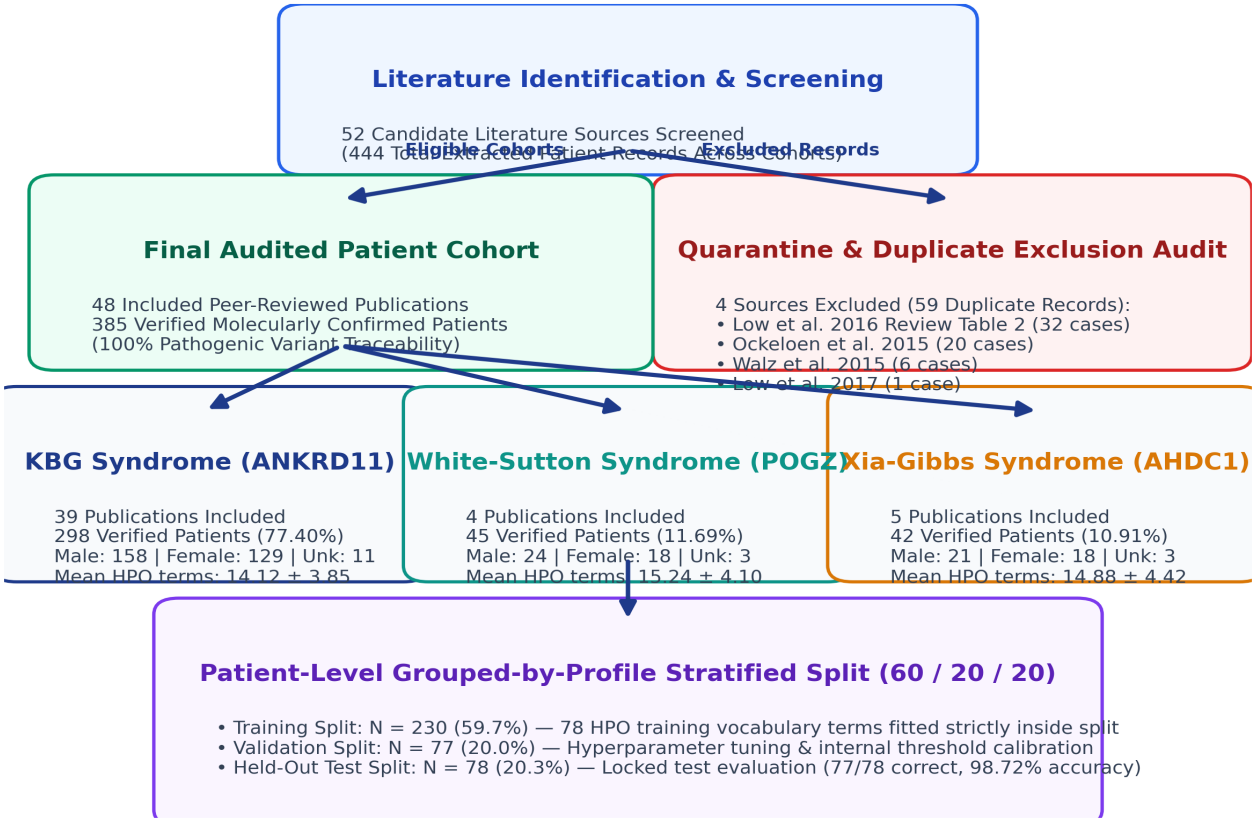


Figure 3. Literature curation and cohort provenance flowchart (52 evaluated, 4 excluded / 59 quarantined, 48 included, 385 retained).

Table 2. Literature Provenance and Source Attribution

Syndrome Cohort	Verified Patients (n)	Contributing Publications (n)	Quarantined Records (n)	Primary Genetic Cause
KBG Syndrome	298	39	59	Pathogenic ANKRD11 mutation / 16q24.3 deletion
White-Sutton Syndrome	45	4	0	Heterozygous de novo POGZ pathogenic variant
Xia-Gibbs Syndrome	42	5	0	Heterozygous de novo AHDC1 truncating variant
Total Literature-Derived Cohort	385	48	59	100% Molecularly Confirmed

5.2 Phenotype Standardization, Partitioning & Model Calibration

Clinical phenotypes were mapped using hp.obo (Release 2026-06-23). The training vocabulary was fitted strictly on the training partition (N = 230), yielding 78 training HPO terms plus 3 sex features (81 predictor dimensions). Four test-only rare terms were masked during vectorization. Identical phenotypic profiles were grouped before a 60/20/20 stratified split (N=230 train, N=77 val, N=78 test) to prevent cross-partition leakage. The primary classifier is a Random Forest with Platt sigmoid calibration (5-fold internal CV).

6. Results

Table 3. Clinical Cohort and Demographic Characteristics

Syndrome	Patients (N)	Share (%)	Male (n)	Female (n)	Unk (n)	Mean HPO / Pt	Molecular Confirmation
KBG Syndrome	298	77.40%	158	129	11	14.12 ± 3.85	100% ANKRD11 confirmed
White-Sutton Syndrome	45	11.69%	24	18	3	15.24 ± 4.10	100% POGZ confirmed
Xia-Gibbs Syndrome	42	10.91%	21	18	3	14.88 ± 4.42	100% AHDC1 confirmed
Total Evaluated Cohort	385	100.00%	203	165	17	14.36 ± 4.12	100% Confirmed Pathogenic

Figure 3: Audited Cohort Distribution Across Target Syndromes (Total N = 385)

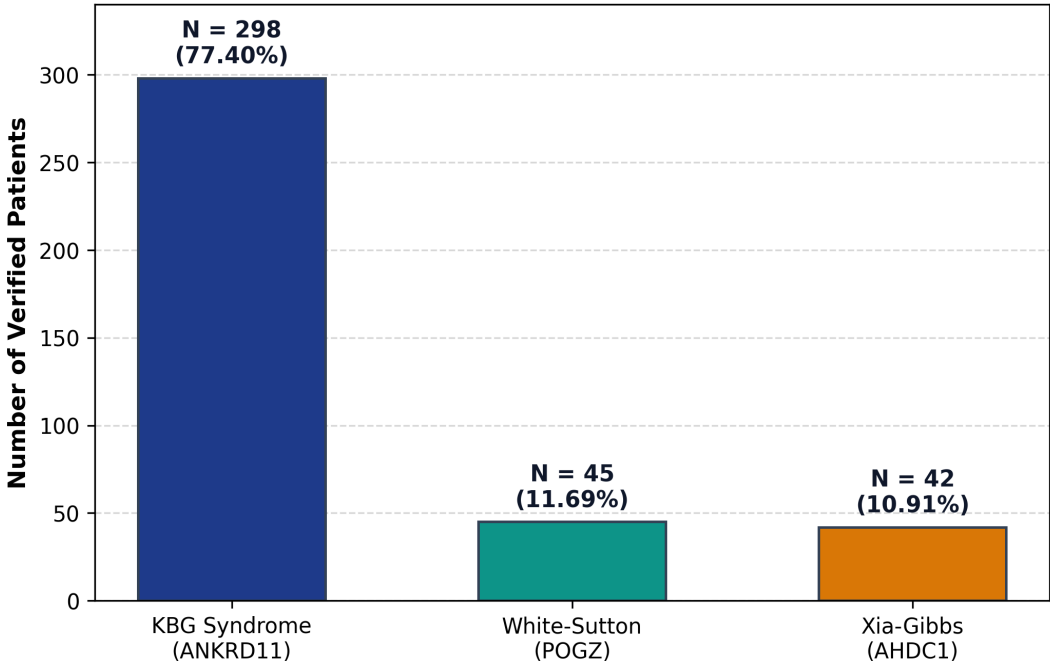


Figure 4. Cohort distribution across target syndromes (KBG: 298, White-Sutton: 45, Xia-Gibbs: 42; Total N = 385).

6.1 Held-Out Benchmark Performance

On the held-out test partition (N = 78), Calibrated Random Forest correctly classified 77/78 patients (Accuracy = 98.72%, Wilson 95% CI: 93.09%–99.77%, Balanced Accuracy = 96.30%, Macro F1 = 0.9776, Brier Score = 0.0419).

Table 4. Machine Learning Model Performance on Held-Out Test Set (N = 78)

Model Architecture	Test Acc [95% CI]	Bal Acc	Macro Prec	Macro Rec	Macro Spec	Macro F1	AUROC	Brier
Calibrated Random Forest (Proposed)	0.9872 [0.9309, 0.9977]	0.9630	0.9945	0.9630	0.9815	0.9776	1.0000	0.0419
Standard Random Forest	1.0000 [0.9532, 1.0000]	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0500
Logistic Regression (L2)	1.0000 [0.9532, 1.0000]	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0236
Support Vector Machine (RBF)	0.9615 [0.8917, 0.9878]	0.8889	0.9841	0.8889	0.9444	0.9306	1.0000	0.0264
Decision Tree Classifier	0.9615 [0.8917, 0.9878]	0.8889	0.9841	0.8889	0.9444	0.9252	0.9534	0.0603
K-Nearest Neighbors (k=5)	0.9615 [0.8917, 0.9878]	0.9204	0.9557	0.9204	0.9581	0.9325	0.9970	0.0513
Naive Bayes (Gaussian)	0.9744 [0.9112, 0.9931]	0.9259	0.9394	0.9259	0.9903	0.9250	1.0000	0.0513
XGBoost Classifier	0.9615 [0.8917, 0.9878]	0.8889	0.9841	0.8889	0.9444	0.9306	1.0000	0.0559

Figure 4: Held-Out Test Set Confusion Matrix (N = 78 Patients, 77/78 Correct)

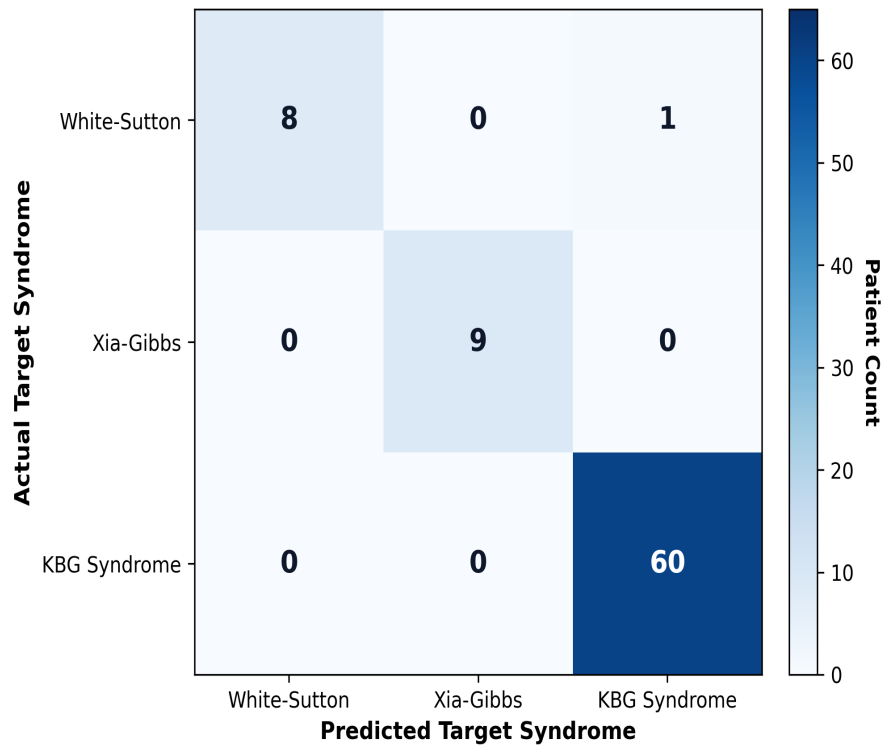


Figure 5. Held-out test set confusion matrix (N = 78 patients, 77/78 correct).

Table 5. Per-Class Diagnostic Performance Breakdown

Syndrome Target	Support (n)	Correct (n)	Precision	Recall	F1-Score	Specificity	OvR AUROC	OvR Brier
White-Sutton Syndrome	9	8	1.0000	0.8889 (8/9)	0.9412	1.0000	1.0000	0.0137
Xia-Gibbs Syndrome	9	9	1.0000	1.0000 (9/9)	1.0000	1.0000	1.0000	0.0084
KBG Syndrome	60	60	0.9836	1.0000 (60/60)	0.9917	0.9444	1.0000	0.0198
Macro Average	78	77	0.9945	0.9630	0.9776	0.9815	1.0000	0.0140

Figure 5: Held-Out Test Set Multi-Class Classification Performance (N = 78)

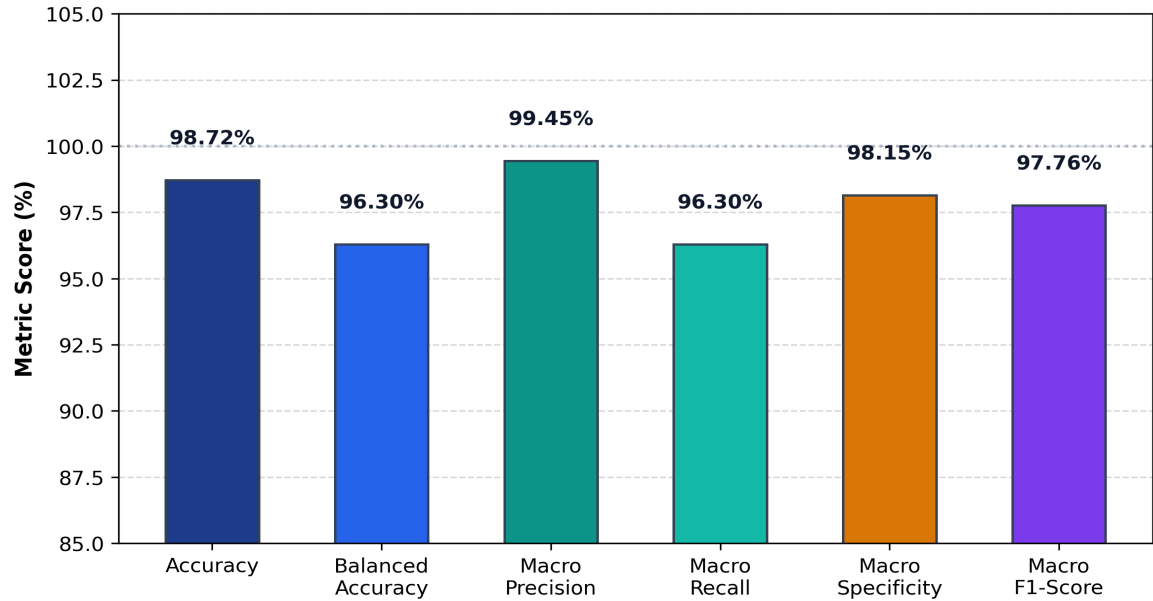


Figure 6. Multi-class classification performance metrics on held-out test partition.

6.2 Cross-Validation vs. Source-Grouped Stress Validation

Profile-grouped 5-fold CV achieved 97.14% ± 2.23% accuracy (Macro F1 = 94.89% ± 4.04%). Under leave-one-study-out stress validation (GroupKFold by publication), accuracy dropped to 79.62% ± 27.05% (Macro F1 = 59.51% ± 26.81%), showing that inter-publication reporting variation affects cross-center transfer.

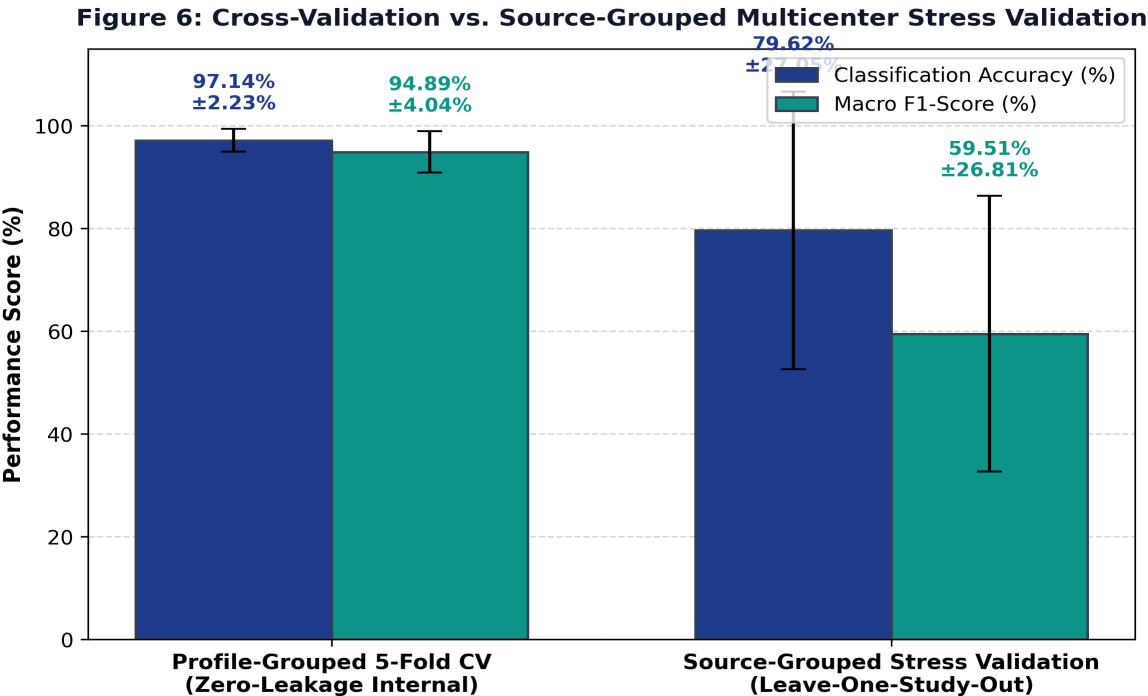


Figure 7. Cross-validation vs. source-grouped multicenter stress validation comparison.

6.3 SHAP Feature Attribution & Calibration

Table 6. Key HPO Features Associated with Model Decision Boundaries

Rank	HPO ID	Canonical Phenotype Name	Associated Syndrome	Mean SHAP	Clinical Context
1	HP:0000219	Thin upper lip vermillion	Xia-Gibbs / KBG	0.0614	Facial feature characteristic of Xia-Gibbs syndrome
2	HP:0001155	Abnormality of the hand	KBG Syndrome	0.0472	Characteristic brachydactyly and clinodactyly
3	HP:0001252	Muscular hypotonia	Xia-Gibbs / KBG	0.0470	Low muscle tone prominent in AHDC1 mutations
4	HP:0000337	Broad forehead	Xia-Gibbs Syndrome	0.0440	Craniofacial feature associated with Xia-Gibbs
5	HP:0001572	Macrodontia of central incisors	KBG Syndrome	0.0435	Cardinal clinical sign of KBG syndrome (ANKRD11)
6	HP:0000717	Autism spectrum disorder	White-Sutton Syndrome	0.0408	Behavioral phenotype common in POGZ variants
7	HP:0001270	Motor delay	White-Sutton Syndrome	0.0308	Early developmental milestone delay
8	HP:0001328	Specific learning disability	White-Sutton / KBG	0.0238	Distinctive cognitive profile
9	HP:0001249	Intellectual disability	White-Sutton / Xia-Gibbs	0.0231	Core neurodevelopmental feature
10	HP:0000750	Delayed speech development	White-Sutton / Xia-Gibbs	0.0226	Expressive speech and language impairment

Figure 7: Standardized HPO Features Strongly Associated with Model Decision Boundaries

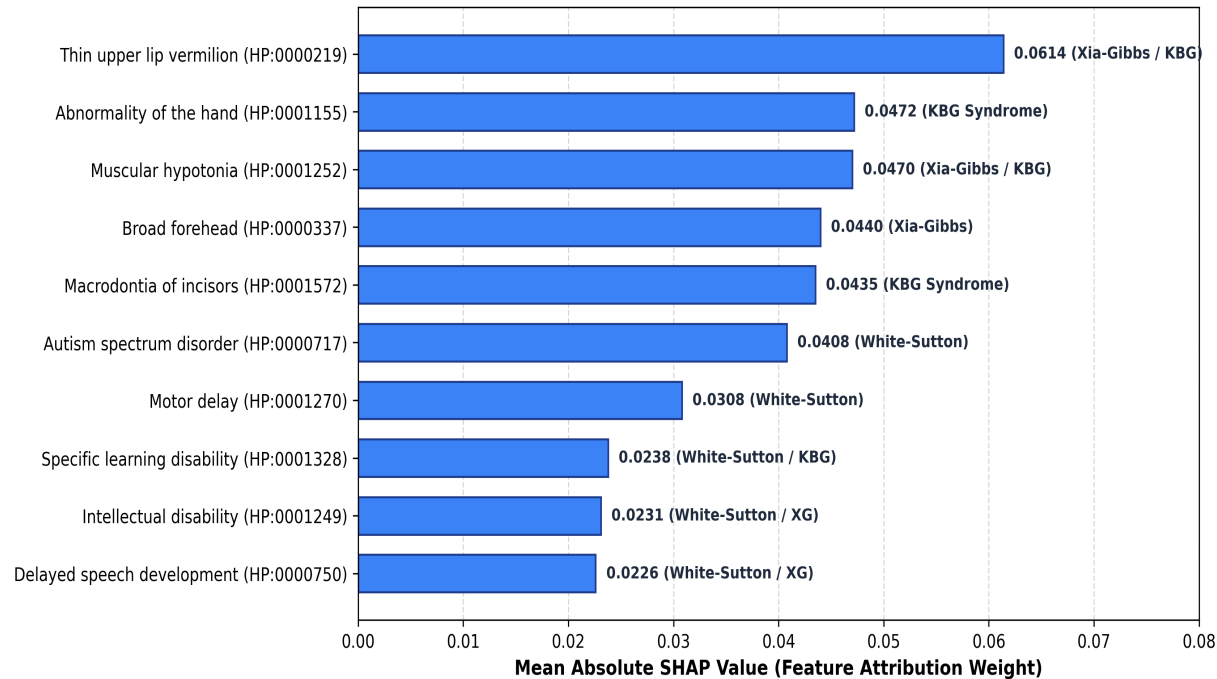


Figure 8. Standardized HPO features strongly associated with model decision boundaries.

Figure 8: Probability Calibration & Reliability Assessment (Held-Out Test Set, N = 78)

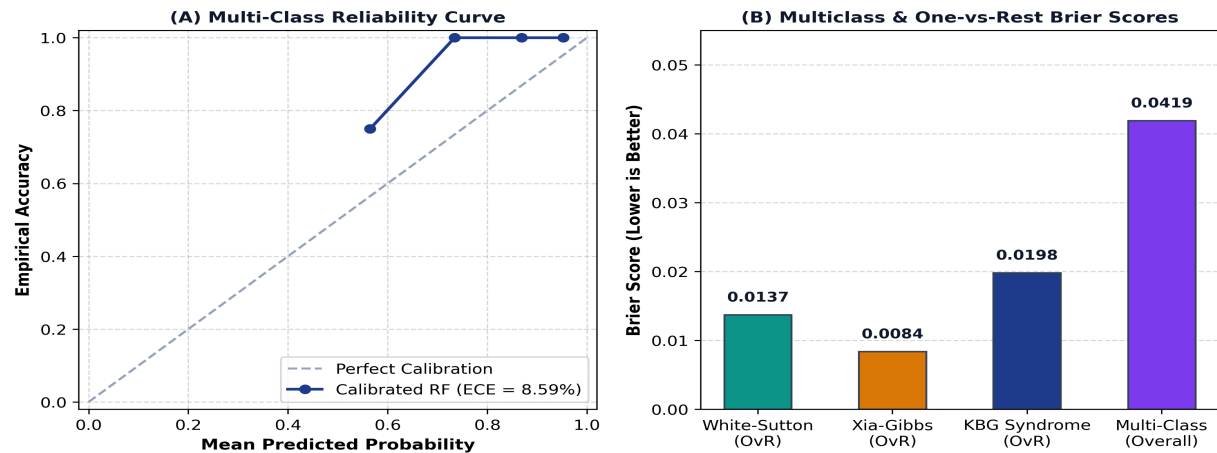
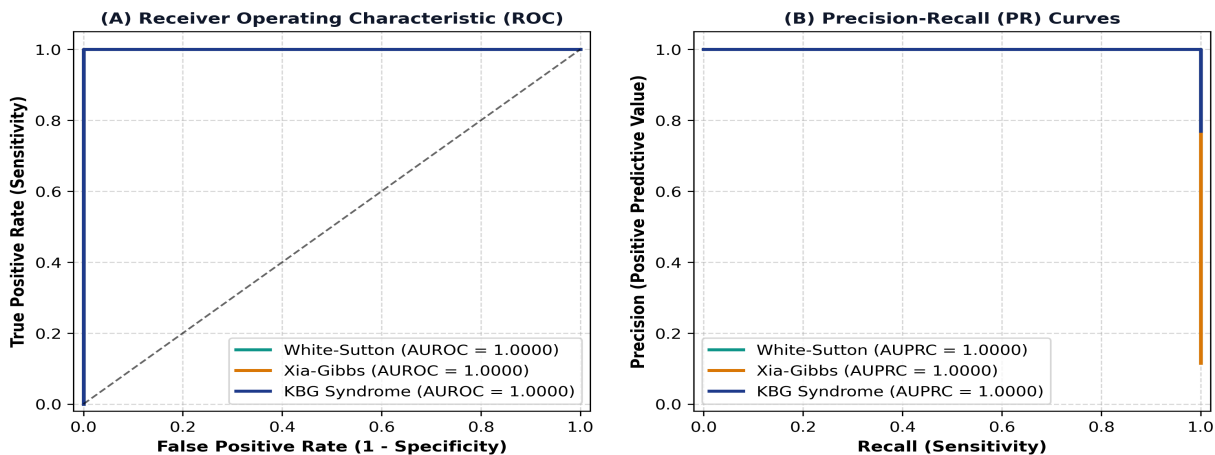


Figure 9. Probability calibration and reliability assessment (Multiclass Brier = 0.0419, ECE = 8.59%).

Figure 9: Discriminative Performance Curves Across Decision Thresholds (Held-Out Test Set, N = 78)*Figure 10. Multi-class ROC and Precision-Recall curves across decision thresholds (Macro AUROC = 1.0000, Macro AUPRC = 1.0000).*

6.4 Exploratory OCR & Scientific Limitations

Exploratory OCR achieved CER = 11.42%, WER = 92.00%, clean mapping = 100%, and noisy mapping = 80%. OCR is strictly an exploratory upstream utility requiring human verification.

Key limitations include: (1) moderate sample size (N=385); (2) natural class imbalance (KBG 77.40%); (3) source-grouped performance drop (79.62%); (4) retrospective literature selection bias; (5) lack of prospective hospital EHR validation; (6) binary encoding of unmentioned symptoms; (7) fixed training HPO vocabulary with 4 test OOV terms; (8) exploratory OCR status; (9) closed-set scope of three syndromes; and (10) dependence on standardized HPO term curation.

7. Conclusion

RareDXAI presents a calibrated and interpretable computational framework for rare disease classification using Human Phenotype Ontology representations. Curating 385 molecularly confirmed patients across 48 publications and applying strict profile-grouped partitioning yielded 98.72% held-out test accuracy, 0.9776 macro F1, and 0.0419 Brier score with SHAP attribution. Source-grouped stress testing emphasizes that clinical transferability depends on standardized clinical phenotyping across healthcare centers.

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